

Entropy: Process Discovery Experiment Report

Compiled at 9:25am on Wednesday 10th December, 2025

1 Introduction

This report summarizes the setup and results of a process discovery experiment conducted using the *Entropy* tool. The following section outlines the discovery techniques used and their configurations in the experiment. Section 3 specifies the evaluation criteria used to assess the quality of the discovered models and to compare the performance of the discovery techniques. Finally, Section 4 presents the experimental results.

2 Discovery

The command line used to execute the experiment is listed below:

```
-mspd -el "BPI_2013.xes" m1 -p 10 -mxItr 20 -t 5000 -mms 500 -pfs 200 -ebrm  
U - optm DFFA
```

Table 1 summarizes the key characteristics of the experimental setup used for performance comparison.

Table 1: Characteristics of the experiment

Parameter	Value
Population size	10
Iterations	20
Time limit (s)	5000
Pareto front size	200
Maximum model size	500
DCI grid siz	4
Optimized models	DFFA
Entropic relevance background model	U
Event log	BPI_2013.xes

These discovery techniques are used in the experiment:

- DE [4];

3 Quality

The quality of the discovered models is assessed using two criteria: *entropic relevance*, which captures stochastic precision and recall [2], and model *size*, defined as the number of nodes and edges, which reflects model simplicity.

The performance of the discovery algorithms is evaluated using two aggregate measures: *dominance count* and the *Diversity Comparison Indicator* (DCI) [3]. Dominance count quantifies how many models discovered by a given algorithm dominate those from other algorithms in the global Pareto front. DCI measures the diversity of the discovered models by analyzing their distribution in the objective space. The details are provided in [1];

4 Results

Figure 1 presents the dominant discovered models with respect to entropic relevance and model size, where lower values are preferred for both criteria. Specifically, Fig. 1a, Fig. 1b, and Fig. 1c show the discovered deterministic frequency finite automata (DFFAs), stochastic deterministic finite automata (SDAGs), and directly-follows graphs (DFGs), respectively. The SDAG models are directly optimized, while the DFFA and DFG models are derived through translations of the optimized SDAGs. Table 5 in Appendix A lists the characteristics of all the discovered models.

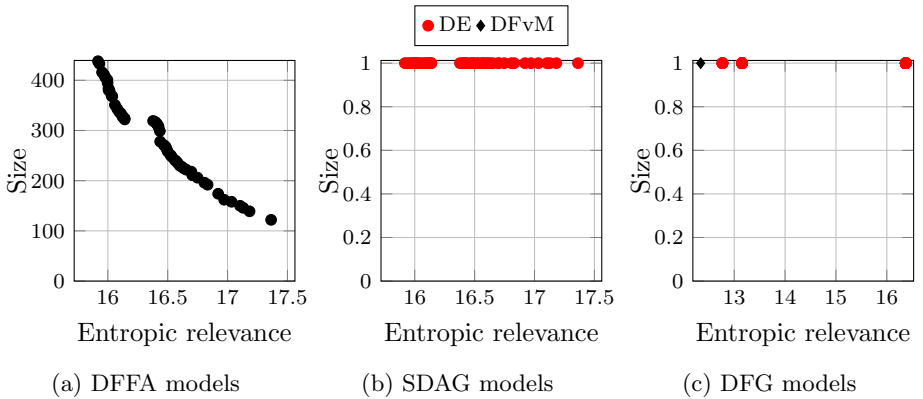


Figure 1: Discovered models plotted by size and entropic relevance for each representation type.

Table 2 and Table 3 summarize the analysis of the diversity of the models and the global Pareto front (dominance count), respectively. Table 4 reports the execution time (in iterations per second) for each input event log and method.

Table 2: Pareto dominance count comparison

Event log	DE(%)	Total
BPI_2013.xes	100.00	6 100.00%

Table 3: DCI comparison (%)

Event log	DE(%)
BPI_2013.xes	61.25

Table 4: Execution time comparison (iterations per second)

Event log	DE
BPI_2013.xes	4.01

References

- [1] Hootan Zhian, Rajkumar Buyya, and Artem Polyvyanyy: *Multi-Objective Metaheuristics for Effective and Efficient Stochastic Process Discovery*. Proceedings of the 23rd International Conference on Business Process Management (BPM) Seville, Spain, August 31–September 5, 2025.
- [2] Hanan Alkhamash, Artem Polyvyanyy, Alistair Moffat, Luciano García-Bañuelos: *Entropic Relevance: A Mechanism for Measuring Stochastic Process Models Discovered From Event Data*. Inf. Syst. 107: 101922 (2022)
- [3] Miqing Li, Shengxiang Yang, Xiaohui Liu: *Diversity Comparison of Pareto Front Approximations in Many-Objective Optimization*. IEEE Trans. Cybern. 44(12): 2568–2584 (2014) Sander J.J. Leemans, Erik Poppe, Moe T. Wynn: *Directly Follows-Based Process Mining: Exploration & a Case Study*. In: ICPM, pp. 25–32 (2019)
- [4] Rainer Martin Storn, Kenneth Price: *Differential evolution - A simple and efficient adaptive scheme for global optimization over continuous spaces*. Tech. Rep. TR-95-012, International Computer Science Institute, 1947 Center Street, Berkeley (1995)

A Model Characteristics

Table 5 summarizes the quality characteristics of discovered models produced by the selected algorithms.

Table 5: Model characteristics: size and entropic relevance

Method	Type	Name	Visualization	Size	Entropic relevance
DE	DFFA	Candidate1	x	241.0	16.559
DE	DFFA	Candidate2	x	307.0	16.423
DE	DFFA	Candidate3	x	343.0	16.079
DE	DFFA	Candidate4	x	258.0	16.500
DE	DFFA	Candidate5	x	235.0	16.584
DE	DFFA	Candidate6	x	230.0	16.601
DE	DFFA	Candidate7	x	315.0	16.403
DE	DFFA	Candidate8	x	218.0	16.697
DE	DFFA	Candidate9	x	222.0	16.654
DE	DFFA	Candidate10	x	268.0	16.481
DE	DFFA	Candidate11	x	192.0	16.833
DE	DFFA	Candidate12	x	317.0	16.395
DE	DFFA	Candidate13	x	158.0	17.033
DE	DFFA	Candidate14	x	150.0	17.104
DE	DFFA	Candidate15	x	139.0	17.184
DE	DFFA	Candidate16	x	319.0	16.378
DE	DFFA	Candidate17	x	146.0	17.130
DE	DFFA	Candidate18	x	225.0	16.632
DE	DFFA	Candidate19	x	278.0	16.437
DE	DFFA	Candidate20	x	264.0	16.489
DE	DFFA	Candidate21	x	322.0	16.141
DE	DFFA	Candidate22	x	380.0	16.007
DE	DFFA	Candidate23	x	174.0	16.922
DE	DFFA	Candidate24	x	122.0	17.364
DE	DFFA	Candidate25	x	239.0	16.570
DE	DFFA	Candidate26	x	299.0	16.434
DE	DFFA	Candidate27	x	337.0	16.096
DE	DFFA	Candidate28	x	395.0	15.998
DE	DFFA	Candidate29	x	438.0	15.918
DE	DFFA	Candidate30	x	196.0	16.807
DE	DFFA	Candidate31	x	206.0	16.749
DE	DFFA	Candidate32	x	251.0	16.523
DE	DFFA	Candidate33	x	329.0	16.122
DE	DFFA	Candidate34	x	384.0	16.005
DE	DFFA	Candidate35	x	162.0	16.972
DE	DFFA	Candidate36	x	211.0	16.705
DE	DFFA	Candidate37	x	248.0	16.533
DE	DFFA	Candidate38	x	312.0	16.414
DE	DFFA	Candidate39	x	379.0	16.014
DE	DFFA	Candidate40	x	228.0	16.616
DE	DFFA	Candidate41	x	368.0	16.037
DE	DFFA	Candidate42	x	406.0	15.978
DE	DFFA	Candidate43	x	433.0	15.930

Method	Type	Name	Visualization	Size	Entropic relevance
DE	DFFA	Candidate44	x	309.0	16.421
DE	DFFA	Candidate45	x	324.0	16.140
DE	DFFA	Candidate46	x	351.0	16.060
DE	DFFA	Candidate47	x	416.0	15.952
DE	DFFA	Candidate48	x	334.0	16.108
DE	DFFA	Candidate49	x	369.0	16.032
DE	DFFA	Candidate50	x	271.0	16.468
DE	DFFA	Candidate51	x	327.0	16.124
DE	DFFA	Candidate52	x	411.0	15.969
DE	DFFA	Candidate53	x	401.0	15.996
DE	SDAG	Candidate1	x	1.0	16.556
DE	SDAG	Candidate2	x	1.0	16.422
DE	SDAG	Candidate3	x	1.0	16.077
DE	SDAG	Candidate4	x	1.0	16.497
DE	SDAG	Candidate5	x	1.0	16.582
DE	SDAG	Candidate6	x	1.0	16.599
DE	SDAG	Candidate7	x	1.0	16.402
DE	SDAG	Candidate8	x	1.0	16.695
DE	SDAG	Candidate9	x	1.0	16.652
DE	SDAG	Candidate10	x	1.0	16.478
DE	SDAG	Candidate11	x	1.0	16.831
DE	SDAG	Candidate12	x	1.0	16.394
DE	SDAG	Candidate13	x	1.0	17.032
DE	SDAG	Candidate14	x	1.0	17.103
DE	SDAG	Candidate15	x	1.0	17.182
DE	SDAG	Candidate16	x	1.0	16.377
DE	SDAG	Candidate17	x	1.0	17.129
DE	SDAG	Candidate18	x	1.0	16.630
DE	SDAG	Candidate19	x	1.0	16.434
DE	SDAG	Candidate20	x	1.0	16.486
DE	SDAG	Candidate21	x	1.0	16.139
DE	SDAG	Candidate22	x	1.0	16.007
DE	SDAG	Candidate23	x	1.0	16.920
DE	SDAG	Candidate24	x	1.0	17.363
DE	SDAG	Candidate25	x	1.0	16.567
DE	SDAG	Candidate26	x	1.0	16.433
DE	SDAG	Candidate27	x	1.0	16.094
DE	SDAG	Candidate28	x	1.0	15.998
DE	SDAG	Candidate29	x	1.0	15.917
DE	SDAG	Candidate30	x	1.0	16.805
DE	SDAG	Candidate31	x	1.0	16.747
DE	SDAG	Candidate32	x	1.0	16.521
DE	SDAG	Candidate33	x	1.0	16.120
DE	SDAG	Candidate34	x	1.0	16.005

Method	Type	Name	Visualization	Size	Entropic relevance
DE	SDAG	Candidate35	x	1.0	16.970
DE	SDAG	Candidate36	x	1.0	16.703
DE	SDAG	Candidate37	x	1.0	16.530
DE	SDAG	Candidate38	x	1.0	16.413
DE	SDAG	Candidate39	x	1.0	16.013
DE	SDAG	Candidate40	x	1.0	16.614
DE	SDAG	Candidate41	x	1.0	16.035
DE	SDAG	Candidate42	x	1.0	15.977
DE	SDAG	Candidate43	x	1.0	15.930
DE	SDAG	Candidate44	x	1.0	16.420
DE	SDAG	Candidate45	x	1.0	16.139
DE	SDAG	Candidate46	x	1.0	16.058
DE	SDAG	Candidate47	x	1.0	15.951
DE	SDAG	Candidate48	x	1.0	16.107
DE	SDAG	Candidate49	x	1.0	16.030
DE	SDAG	Candidate50	x	1.0	16.465
DE	SDAG	Candidate51	x	1.0	16.122
DE	SDAG	Candidate52	x	1.0	15.967
DE	SDAG	Candidate53	x	1.0	15.994
DE	DFG	Candidate1	x	1.0	13.153
DE	DFG	Candidate2	x	1.0	13.153
DE	DFG	Candidate3	x	1.0	16.373
DE	DFG	Candidate4	x	1.0	13.153
DE	DFG	Candidate5	x	1.0	13.153
DE	DFG	Candidate6	x	1.0	13.153
DE	DFG	Candidate7	x	1.0	16.373
DE	DFG	Candidate8	x	1.0	13.153
DE	DFG	Candidate9	x	1.0	13.153
DE	DFG	Candidate10	x	1.0	13.153
DE	DFG	Candidate11	x	1.0	13.153
DE	DFG	Candidate12	x	1.0	16.373
DE	DFG	Candidate13	x	1.0	12.769
DE	DFG	Candidate14	x	1.0	12.769
DE	DFG	Candidate15	x	1.0	12.769
DE	DFG	Candidate16	x	1.0	16.373
DE	DFG	Candidate17	x	1.0	12.769
DE	DFG	Candidate18	x	1.0	13.153
DE	DFG	Candidate19	x	1.0	13.153
DE	DFG	Candidate20	x	1.0	13.153
DE	DFG	Candidate21	x	1.0	16.373
DE	DFG	Candidate22	x	1.0	16.373
DE	DFG	Candidate23	x	1.0	13.153
DE	DFG	Candidate24	x	1.0	12.769
DE	DFG	Candidate25	x	1.0	13.153

Method	Type	Name	Visualization	Size	Entropic relevance
DE	DFG	Candidate26	x	1.0	13.153
DE	DFG	Candidate27	x	1.0	16.373
DE	DFG	Candidate28	x	1.0	16.373
DE	DFG	Candidate29	x	1.0	16.373
DE	DFG	Candidate30	x	1.0	13.153
DE	DFG	Candidate31	x	1.0	13.153
DE	DFG	Candidate32	x	1.0	13.153
DE	DFG	Candidate33	x	1.0	16.373
DE	DFG	Candidate34	x	1.0	16.373
DE	DFG	Candidate35	x	1.0	12.769
DE	DFG	Candidate36	x	1.0	13.153
DE	DFG	Candidate37	x	1.0	13.153
DE	DFG	Candidate38	x	1.0	16.373
DE	DFG	Candidate39	x	1.0	16.373
DE	DFG	Candidate40	x	1.0	13.153
DE	DFG	Candidate41	x	1.0	16.373
DE	DFG	Candidate42	x	1.0	16.373
DE	DFG	Candidate43	x	1.0	16.373
DE	DFG	Candidate44	x	1.0	16.373
DE	DFG	Candidate45	x	1.0	16.373
DE	DFG	Candidate46	x	1.0	16.373
DE	DFG	Candidate47	x	1.0	16.373
DE	DFG	Candidate48	x	1.0	16.373
DE	DFG	Candidate49	x	1.0	16.373
DE	DFG	Candidate50	x	1.0	13.153
DE	DFG	Candidate51	x	1.0	16.373
DE	DFG	Candidate52	x	1.0	16.373
DE	DFG	Candidate53	x	1.0	16.373